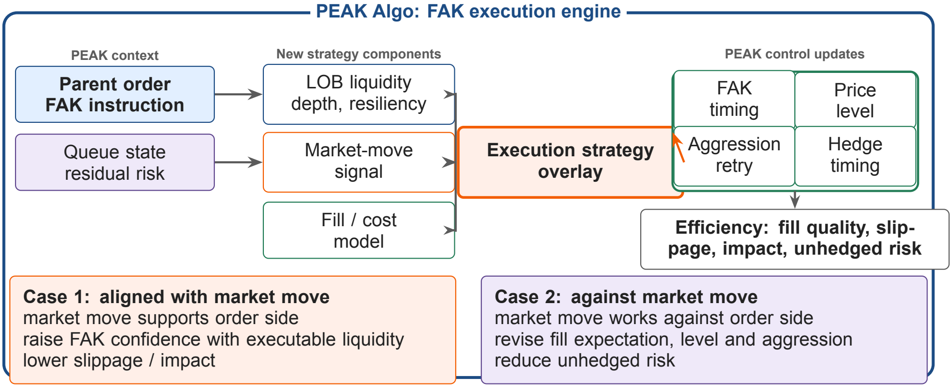
Liquidity-Aware Smart Execution

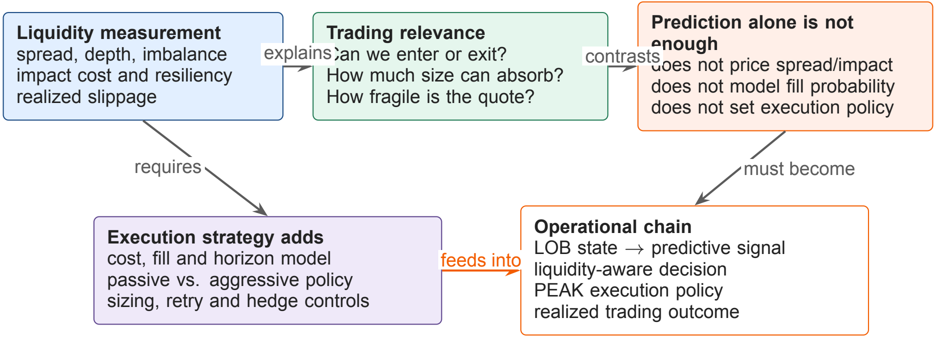
Microstructure Strategy for FMD and Peak Algo

Quant team | June 2026

FAK Order Execution in PEAK Algo



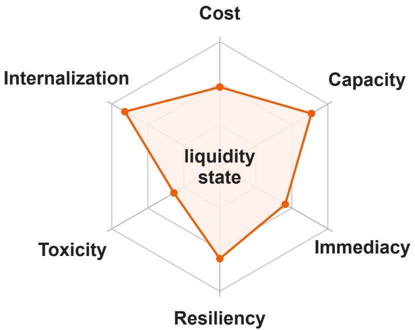
Liquidity Turns Prediction into Execution Strategy



Sustainable execution P&L requires both prediction and a measurable liquidity / execution model.

Liquidity is the True State Variable

$\mathcal{L}_t = \{$cost, capacity, immediacy, resiliency, toxicity, internalization$\}$



How to read the state

- **High capacity + fast resiliency:** raise FAK confidence.
- **High cost or toxicity:** revise fill expectation and aggression.
- **High internalization:** match internally before external routing.
- **Weak state:** delay, slice smaller, or hedge earlier.

Design principle

Price is the observation. Liquidity is the control state.

Source: Lehalle and Laruelle; Cont and de Larrard; Bacry et al.; LOBFrame / LOB benchmark studies.

0.01 bp as Attributed Efficiency Opportunity

Addressable efficiency opportunity

90M

potential annual efficiency from 0.01 bp saving

- 2025 flow: 45tn.
- Average duration: 2 years.
- $45tn \times 2 \times 0.01bp \approx 90M$.

Replay · Shadow · Pilot · Scale

Where the 0.01 bp opportunity comes from

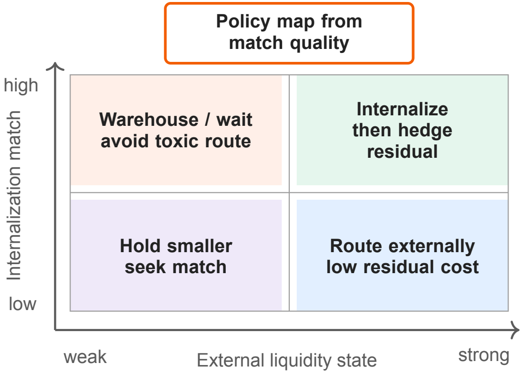
Liquidity timing — depth + resiliency windows

Toxicity control — hedge leakage + bad fills

Internalization — avoid spread + external impact

Attribution ledger: slippage, impact, internalization uplift

Internalization: Match Before Route



Matching inputs

- Client orders, RFQ, axes and inventory.
- Side, tenor, size and urgency.
- Hedge cost, limits and residual risk.

Execution policy

- Internalize when match quality is high.
- Route only the measured residual.
- Measure savings vs. external route cost.